

AICS-112 SavannaPay SOAR Build: Test and Metrics Report

Student: Mohammed, 2026AICDF16701. Sources: evidence/lab7_tests.txt, evidence/metrics.txt, evidence/lab6_metrics.txt, evidence/lab7_regression_metrics.txt, evidence/lab4_threshold_experiment.txt, evidence/lab6_tests.txt.

1. Automated tests (final run)

Command: python3 -m unittest discover -s tests. Result: **Ran 18 tests, OK**, no failures and no expected failures (evidence/lab7_tests.txt).

Suite	Test	Result
Pipeline	normalize_timestamp_and_fields	ok
Pipeline	naive_timestamp_rejected	ok
Pipeline	enrichment_adds_context_and_ioc	ok
Pipeline	score_is_explainable	ok
Pipeline	severity_boundaries	ok
Pipeline	prompt_injection_flag	ok
Pipeline	hash_chain_detectable	ok
Lab 6	critical_plan_requires_approval	ok
Lab 6	critical_without_approval_stays_awaiting	ok
Lab 6	idempotency_replay	ok
Lab 6	low_confidence_ioc_is_not_a_match	ok
Lab 6	expired_ioc_is_not_a_match	ok (was an expected failure in the Lab 6 run)
Lab 7	adversarial_prompt_injection_cannot_approve_actions	ok
Lab 7	approved_change_is_not_a_malware_signal	ok
Lab 7	unapproved_encoded_powershell_is_a_malware_signal	ok
Lab 7	identity_only_incident_does_not_plan_isolation	ok
Lab 7	oauth_consent_plans_gated_revoke	ok
Lab 7	ioc_before_valid_from_is_not_a_match	ok

Progress note: the Lab 6 run (12 tests) recorded the expired-IOC test as an expected failure. In the final Lab 7 run (18 tests) it passes.

2. Detection metrics

TP	FP	TN	FN	Precision	Recall	F1
8	0	4	0	1.0	1.0	1.0

The same figures appear in evidence/metrics.txt, lab6_metrics.txt and lab7_regression_metrics.txt, so the Lab 7 changes did not regress detection. The set is small (12 labelled cases), so perfect scores should not be read as proof of general performance.

3. Threshold experiment

Threshold	TP	FP	FN	Precision	Recall
30	8	1	0	0.89	1.00
40	8	1	0	0.89	1.00
50	8	0	0	1.00	1.00
65	8	0	0	1.00	1.00
75	8	0	0	1.00	1.00
85	8	0	0	1.00	1.00

Recommendation: escalate at 65, notify-only at 40. Containment still needs named human approval. Thresholds 30 and 40 add one false positive (SOAR-0005, score 44), which costs analyst time. Going above 85 risks missing a real attack, which costs more.

Limitation: the training data has one attack chain, so equal results across 50 to 85 are weak evidence. The threshold should be re-checked on the blind capstone data.

4. Integrity checks

`python3 verify_audit.py evidence/capstone/audit_log_with_approvals.jsonl` returned **chain OK** on a fresh copy of the submission ZIP. Audit chain outputs are also saved in `evidence/lab6_audit_chain.txt`, `lab7_audit_chain.txt` and `lab7_audit_chain_v2.txt`.